

1 Introduction: What is Free Fall?

How could you study motion if you didn't have accurate clocks, much less video analysis or electronic measurement equipment?

This was the problem faced by Italian physicist Galileo Galilei in the late sixteenth century. He studied mechanics, beginning with the simplest cases. One of those simple cases is *free fall*. Free fall is what happens when you hold a stone up in the air, then let it go. People knew that the stone would get faster and faster as it fell, but disagreed on the details.

Some people thought that stone's instantaneous velocity would be proportional to how far it fell. (From here on in this course, you can assume that words like "speed" and "velocity" mean the *instantaneous* speed or velocity, not the average speed or velocity, unless we specifically say it's an average.)

If the velocity of the stone is v_y and the distance it has fallen is Δy , then this hypothesis is

$$v_y \stackrel{?}{\propto} \Delta y \quad (1.1)$$

This is a *proportionality*. It is the same as $v_y = k \cdot \Delta y$, where k is a constant whose value isn't important to the hypothesis. This hypothesis says that if an object is moving at velocity -3 m/s after falling -1 m, will be moving at velocity -6 m/s after falling -2 m and velocity -45 m/s after falling -15 m, etc. (We use negatives for all these values because we are defining the y coordinate to increase in the upward direction, so $v_y < 0$ and $\Delta y < 0$ when a stone falls.) The precise numbers aren't important; the relationship between velocity and distance fallen is. According to this hypothesis, multiplying the distance fallen by some number multiplies the velocity by the same number.

Others believed the stone's velocity would be proportional to the *time* the object had been falling,

$$v_y \stackrel{?}{\propto} t. \quad (1.2)$$

Imagine you hold a stone up and drop it. It hits the ground at some velocity. Suppose you want it to hit the ground twice as fast. The first hypothesis, $v_y \propto \Delta y$, says that you need to hold the stone twice as high and drop it. The second, $v_y \propto t$, says that you need to hold the stone high enough that it will take twice as long to hit the ground.

Galileo wanted to know which (if either) of these hypotheses were correct.

Problem 1.1. Are the two hypotheses different?

Maybe if you hold a stone twice as high and drop it, it will hit the ground twice as fast. That would confirm $v_y \propto \Delta y$. But what if that stone also takes twice as long to fall? That would confirm $v_y \propto t$ as well. Then *both* hypotheses would be right.

If two hypotheses make all the same predictions, then scientists consider them to be different ways of stating the same hypothesis.

Do the hypotheses $v_y \propto \Delta y$ and $v_y \propto t$ make different predictions? If so, give an example. If not, prove they are the same.

Solution: Let's imagine letting a stone fall a distance Δy_1 in time t_1 . At this point, the velocity of the stone is v_{y1} . We allow the stone to continue falling until it has gone a distance $2\Delta y_1$.

If $v_y \propto \Delta y$ is correct, then the velocity of the stone is $2v_{y1}$. Let's think about what the time is.

The stone has fallen through two segments, both of length Δy_1 . The stone fell faster in the second segment than in the first, though, because the stone is always getting faster as it falls. That means the stone took less time for the second segment than for the first. The stone will fall the distance $2\Delta y_1$ in a time $t_2 < 2t_1$. The stone will have to continue falling, and continue getting faster, if it falls until time $2t_1$.

The hypothesis $v_y \propto t$ predicts that at time $2t_1$, the velocity of the stone is $2v_{y1}$. The hypothesis $v_y \propto \Delta y$ predicts that at time $2t_1$, the stone will be going even faster than that.

These are different predictions, so the hypotheses are different. □

Galileo wanted to design an experiment, but he had no good way to measure velocity, and measuring times was difficult as well. A stone dropped from head height hits the ground in around half a second, which is so short that Galileo's primitive clocks (based on the flow of water) couldn't measure it accurately. Galileo needed a way to slow the motion down before he could make quantitative statements about free fall.

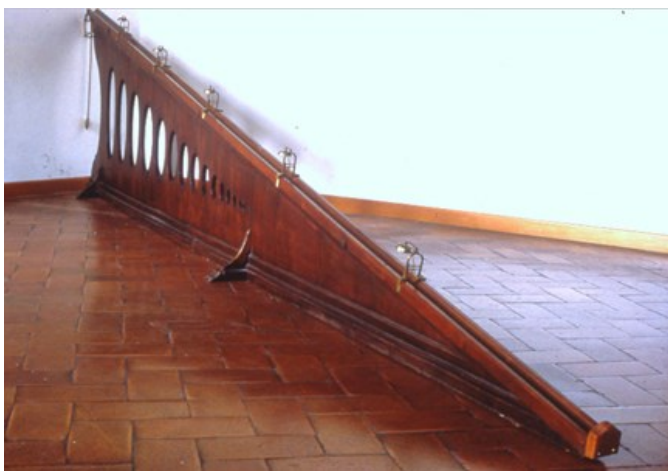
Exercises

- 1.1. Explain in your own words why we want to do a calculation to test whether $v_y \propto \Delta y$ and $v_y \propto t$ are different.
- 1.2. Suppose $v_y \propto \Delta y$. You drop a stone and find it travels at 5 m/s after falling 2 m. How fast is it going after falling 12 m?
- 1.3. Suppose $v_y \propto t$. You drop a stone and find it travels at 5 m/s after falling 2 m. After falling 12 m, is it going faster, slower, or the same speed as what you found in the previous exercise?
- 1.4. Suppose $v_y \propto \Delta y$. Can you make an argument that when a stone is released, it should, according to this hypothesis, hover where it was without moving? Why doesn't this same argument also apply to $v_y \propto t$?

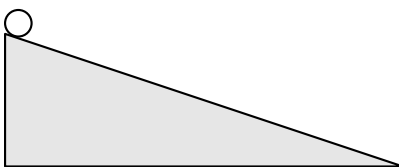
2 Ringing Bells

Galileo decided to study a slower motion than free fall. Instead, he rolled balls down an *inclined plane*, a straight ramp.

Here is a replica inclined plane, similar to the one Galileo used:



Here is a diagram:



A ball on an inclined plane like this will get faster and faster as it rolls down the inclined plane. Galileo assumed this motion obeys the same rule as free fall, either $v_x \propto \Delta x$ or $v_x \propto t$. (We have switched the coordinate to x instead of y because the ramp is mostly horizontal, but there's no hard-and-fast rule in physics about what you call your coordinates.) The constant k in $v_x = k\Delta x$ or $v_x = kt$ would be smaller for the inclined plane than for free fall, but Galileo wasn't trying to determine that constant.

Galileo still needed a way to make measurements on the ball as it rolled. He couldn't measure the velocity directly - there was nothing like a radar gun used today. Instead, he measured the *position* of the ball over time, and used the position versus time to figure out what the velocity must have been.

Even measuring the position over time was challenging due to the difficulty in measuring times, so Galileo came up with an ingenious solution. (Historically, Galileo may or may not have used this solution; it isn't known. We'll analyze it because it is conceptually simple.)

Galileo hung a series of bells over the inclined plane. As the ball rolled down, it rang each bell. Galileo then adjusted the positions of the bells until they rang in a regular rhythm, with the same amount of time between each ring, like the beat of a song.

Humans can tell to good accuracy whether a series of sounds come in a regular pattern or not. This means Galileo was able to get the bells to ring with the nearly the same amount of time between them.

Problem 2.1. Did Galileo space all the bells equally far apart? If not, how did the space between them change as the ball rolled down the ramp?

Solution: The bells are essentially a dot diagram for the ball's motion. In a dot diagram, we draw dots showing the

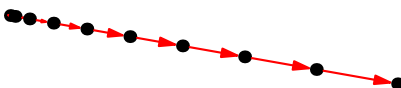
position of an object. The dots are evenly spaced in time. In this case, the bells were in positions that the ball reached in evenly spaced times.

As the ball rolled down the inclined plane, it got faster and faster, so traveled further and further between bells.

Here is a dot diagram, with dots showing the spacing of the bells.



Since dot diagrams don't indicate the direction of motion, we can modify the diagram by drawing arrows from dot to dot.



When we add the arrows, we'll call it a *motion diagram*. The length of each arrow is the displacement between dots, and is proportional to the magnitude of the average velocity of the object between dots. □

If the bells had been spaced evenly far apart, the time between rings would have gotten shorter and shorter as the ball went down the inclined plane.

Exercises

- 2.1. Draw a dot diagram for a ball that begins at the bottom of an inclined plane with a high speed and slows down as it rises. The ball should move to the right.
- 2.2. Add arrows to your dot diagram to turn it into a motion diagram.
- 2.3. Draw a motion diagram for a ball that rolls down an inclined plane, continues on a flat section of track for a short time, then rises up another inclined plane, slowing down before stopping at the top of the second inclined plane.



3 Galileo's Results

Galileo conducted the experiment on an inclined plane many times. He tried inclined planes of different lengths and angles. He also had the bells ring closer together in some experiments and further apart in others. Each time, he

measured the distances between each set of bells.

In these experiments, the distances were all different, but they always followed the same pattern. To describe the pattern, Galileo didn't give the distance between the bells in inches or centimeters. Instead, he reported the distances *relative* to the distance from where the ball was released to the first bell.

For example, if the ball rolled 2 cm to get to the first bell and 6 cm to get to the second bell, what Galileo reported was that the ball rolled three times as far to get to the second bell as it rolled to get to the first.

Although the distances measured in centimeters were all different in different experiments, the relative distances remained the same. For example, if the ball rolled 3 cm to get to the first bell, it rolled 9 cm to get to the second one, again three times as far.

If the distance to the first bell was d , then the distance from the first to second bell was always $3d$.

Additionally, the distance from the second to the third bell was always $5d$, and from the third to fourth bells $7d$. The gaps between later bells would be $9d$, $11d$, $13d$, etc.

The distance between bells increased as Galileo looked further down the inclined plane. However, there was something that stayed constant: the amount that the distance between bells increased each time.

Between the first and second bells, the distance increased from d to $3d$, which is an increase of $2d$. From the second to third bells, the distance increased from $3d$ to $5d$, another increase of $2d$.

When we calculate the change in distance to adjacent bells, like the example in the last paragraph, we call it the *second difference* of the positions (the bells themselves are first differences in the position of the ball). The second difference was the same all the way down the ramp: always $2d$. So although the velocity of the ball was changing, there was still something Galileo could point to that was constant.

This type of motion is called *constant acceleration*. You may also see it called *uniform acceleration* in some places. These mean the same thing.

"Acceleration" means any change in velocity. "Constant" means the acceleration does not change with time. We'll cover more cases of acceleration in class, but for this handout, we'll stick to constant acceleration starting from rest.

Problem 3.1. Draw a motion diagram for an object with constant acceleration, starting from rest and moving horizontally. Make the diagram to scale and label the displacements.

Solution:



□

Although he confirmed the mathematical rule for motion with constant acceleration by studying rolling down an inclined plane, Galileo applied the model to free fall as well. If an object that is dropped near Earth falls a distance d in its first second of falling, it will fall another $3d$ in the next second and $5d$ in the second after that, etc.

This is true for other periods of time besides 1 s. For example, if an object starts from rest and has constant acceleration and travels a distance 1 m in its first 0.2 s, it will travel a distance 3 m in its next 0.2 s, etc.

Exercises

- 3.1. On Earth, any object in free fall will take about 0.45 s to fall one meter, assuming that we can ignore effects like air resistance. How far would the object fall in the time between 0.90 s and 1.35 s after it was dropped?
- 3.2. How long would it take the object to fall all the way to the ground, if it were dropped from a height of 4 m?
- 3.3. Approximately how long do you think it would take the object to fall to the ground if it were dropped from a height of 2 m? You don't need to answer this question exactly yet; we will learn how to do it more precisely by the end of the handout. For now, use what you already understand about constant acceleration to find a reasonable answer that is close to the precise calculation.

4 Rewriting Galileo's Result

Problem 4.1.

Find the *total* distance traveled by an object with constant acceleration, starting from rest, as it goes past the first, second, and third bells. Is there a pattern? Will the pattern continue to hold as long as Galileo's rule does? Explain how you know.

Solution: At the first bell, the distance is d . To get to the second bell, a ball travels an additional $3d$, so the total distance traveled is $d + 3d = 4d$. At the third bell, the total distance traveled is $d + 3d + 5d = 9d$.

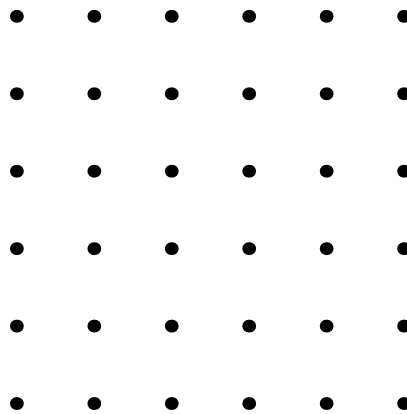
We can continue adding successive odd multiples of d to find the total distance traveled at later and later bells. The results are

bell	incremental distance (d)	total distance (d)
1	1	1
2	3	4
3	5	9
4	7	16
5	9	25
6	11	36

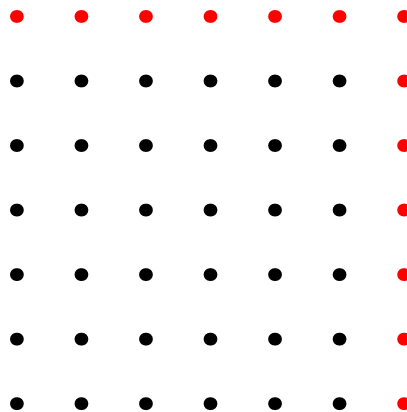
It looks like to get to bell n , the total distance traveled is n^2d .

We can test this pattern by looking at bell 7, then bell 8, etc. However, we have to stop somewhere, and we won't know if the pattern stops holding someplace beyond the last bell we checked.

To see whether the pattern will hold for any number of bells, we visualize the number n^2 as the number of dots in an $n \times n$ grid. Below, we've chosen $n = 6$.



If we want to go to $n = 7$, we can add dots all along the upper and right hand sides of our grid, plus one dot in the top right corner. These added dots are shown in red below.



There are 13 extra dots added: 6 along the right, 6 along the top, and one in the upper right corner.

The next row of our table will therefore be

bell	incremental	distance (d)	total	distance (d)
1		1		1
2		3		4
3		5		9
4		7		16
5		9		25
6		11		36
7		13		49

The same will work for any value of n . If we start with an $n \times n$ grid and want to create an $(n + 1) \times (n + 1)$ grid, we can add n dots along the right hand side, n dots along the top, and one dot in the upper right corner, for $2n + 1$ additional dots. The numbers $2n + 1$ are all the odd numbers, so this is the pattern we identified in our table. It holds for as many rows as we want.

□

Another way we can rewrite Galileo's result is to use algebraic notation. We introduce the variable Δx for the distance the ball has rolled (or Δy for the distance fallen in free fall) and the variable t for the time the ball has been moving, from rest, with constant acceleration. Then we can write Galileo's rule as

$$\Delta x \propto t^2 \quad (4.1)$$

for rolling down the ramp, or

$$\Delta y \propto t^2 \quad (4.2)$$

for free fall.

Problem 4.2. An object in free fall travels 1 m in the first 0.45 s of its fall. How long will it take to fall a total of 2 m?

Solution: The distance fallen, Δy , obeys

$$\Delta y \propto t^2. \quad (4.3)$$

This means that if we have two different distances, Δy_1 and Δy_2 , then

$$\frac{\Delta y_1}{\Delta y_2} = \frac{t_1^2}{t_2^2}. \quad (4.4)$$

If you are less familiar with proportionalities, we can show this via equations. As an equation, $\Delta y = k \cdot t^2$ for some unknown k . Then

$$\frac{\Delta y_1}{\Delta y_2} = \frac{k \cdot t_1^2}{k \cdot t_2^2}. \quad (4.5)$$

The k 's cancel out, so it doesn't matter what k is, and we can determine the ratio $\Delta y_1/\Delta y_2$ using just the proportionality. Solving for t_2 ,

$$t_2 = t_1 \sqrt{\frac{\Delta y_2}{\Delta y_1}}. \quad (4.6)$$

Here, we take $\Delta y_1 = 1$ m and $t_1 = 0.45$ s. We are doubling the distance fallen to $\Delta y_2 = 2$ m, so the time is

$$t_2 = 0.45 \text{ s} \cdot \sqrt{2} = \boxed{0.64 \text{ s}}. \quad (4.7)$$

□

Exercises

- 4.1. How long will it take an object in free fall to fall 20 m? How long will it take to fall 80 m? How long to fall 320 m? How long to fall 1280 m? Assume the object begins from rest.
- 4.2. Explain the pattern in the result to the previous exercise and relate it to the proportionality $\Delta x \propto t^2$.
- 4.3. Show by algebra that we always add $2n + 1$ to go from n^2 to $(n + 1)^2$. How could we have used this fact, instead of the square grid of dots, in problem 4.1?
- 4.4. Suppose you are making a movie at home, and you want to show a hero jumping off a tall cliff. You film the scene by tossing a small action figure off a small replica cliff you made. It hits the ground in less time than your hero would have in real life, so you decide to play the clip in slow motion in the movie. The replica cliff is $1/16$ as tall

as the real cliff you want to depict. By what factor should you slow down the clip before putting it in your movie?

4.5. (optional) Before Galileo, Leonardo da Vinci came up with a similar law for motion during free fall. Leonardo believed that if an object in free fall travels a distance d in the first period of its fall, it will travel a distance $2d$ in the second period, $3d$ in the third period etc. This is like Galileo's rule, but the numbers 1, 3, 5, 7, ... are replaced with 1, 2, 3, 4, ... Show that it is not possible to find an equation for x in terms of t using Leonardo's rule - the rule itself is contradictory. Do this by coming up with an example where Leonardo's rule makes two different predictions for how far an object in free fall should have traveled in a given time.

5 Judging the Hypotheses

We began with two hypotheses for how velocity changes during free fall: $v_y \propto t$ or $v_y \propto \Delta y$.

We now have experimental evidence about the distance traveled: $\Delta y \propto t^2$.

We need to make a connection between the distance-time relationship and the velocity-time relationship if we want to say which hypothesis is supported by the data.

To do that, let's imagine $v_y \propto t$ and ask ourselves what Δy should do, if this is true. To be specific, we'll let

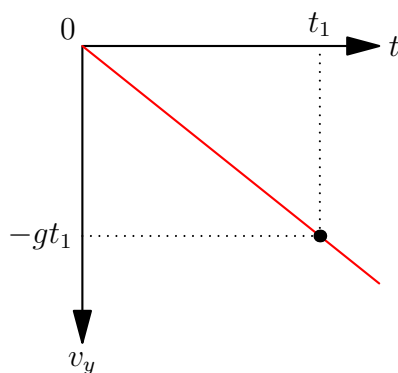
$$v_y = -g \cdot t. \quad (5.1)$$

The minus sign is there because the object falls downward, which we define to be the negative direction. This way, $g > 0$.

The constant g is called the *free fall acceleration*. (It is not a constant of nature; it is the same for all objects at a given place, but different on different planets, and even varies some from place to place on Earth.)

Problem 5.1. Suppose the instantaneous velocity of an object is $v_y = -gt$. Find the displacement Δy of the object after a time t .

Solution: Let's first draw a plot of the velocity over time. This is a straight line with slope $-g$. We'll choose some particular time t_1 and label it on our plot.



Between $t = 0$ and $t = t_1$, the velocity decreases steadily from 0 to $-gt_1$, so the average velocity is $\frac{-gt_1}{2}$. That makes the displacement of the object

$$\Delta y_1 = v_{y;\text{avg}} \cdot t_1 \quad (5.2)$$

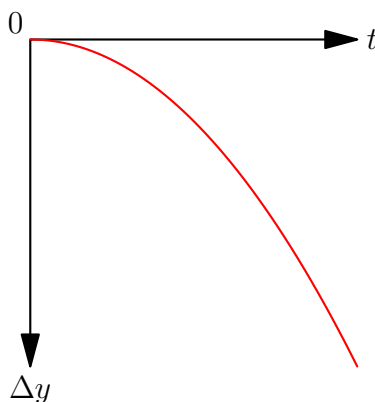
$$= \frac{-gt_1}{2} \cdot t_1 \quad (5.3)$$

$$= \frac{-gt_1^2}{2} \quad (5.4)$$

The same analysis holds at any time, so the position is

$$\boxed{\Delta y = \frac{-gt^2}{2}}. \quad (5.5)$$

A plot of distance fallen over time is a *parabola*.



□

The hypothesis $v_y \propto t$ reproduces our result of $\Delta y \propto t^2$. In other words, Galileo's experimental results are exactly what we should expect to see if objects gain the same amount of velocity every second as they fall or roll down an inclined plane. This is the constant acceleration model of free fall.

The idea that objects gain the same velocity for every meter they fall is quite different. We won't work out the mathematics behind this hypothesis in detail, but it leads to a type of relationship between displacement and time called *exponential*. There can be some circumstances where this model makes accurate predictions of motion, but not for free fall, rolling down an inclined plane, or other motions that we'll study this week.

6 References and Notes

Although most sources give credit to Galileo for experimentally testing different laws of accelerated motion, he was not the first person to model velocity as proportional to time in free fall. Nicole Oresme stated that hypothesis

and drew diagrams that allowed him to calculate the distance fallen in a given time in *On the Latitude of Forms* about 200 years before Galileo. For more, see:

Clagett, Marshall *Nicole Oresme and the Medieval Geometry of Qualities and Motions: A Treatise on the Uniformity and Difformity of Intensities Known as Tractatus de configurationibus qualitatum et motuum*. Madison: University of Wisconsin Press. 1970.

7 Credits

inclined plane replica: Museo Galileo, <https://catalogue.museogalileo.it/object/InclinedPlane.html>

8 Answers to exercises

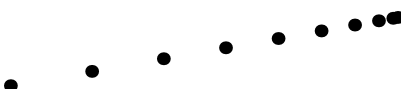
1.1: If $v_y \propto \Delta y$ and $v_y \propto t$ wind up making the same predictions about what will happen in any experiment, then any experiment will either confirm both hypotheses' predictions or go against both hypotheses predictions. Either way, we can't say that one hypothesis is right and the other wrong.

1.2: $12 \text{ m} \cdot \frac{5 \text{ m/s}}{2 \text{ m}} = 30 \text{ m/s}$

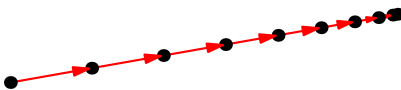
1.3: Slower. The stone is always going faster and faster, so it will take less than 6 times as long to go 12 m as it took to go 2 m. Since it gets less than six times as long to speed up, it will go less than 6 times as fast.

1.4: If $v_y \propto \Delta y$, then when the stone is released, $\Delta y = 0$ and $v_y = 0$, so the stone has no velocity. That's correct: the stone is released from rest. However, if we go one tiny moment forward in time, the stone still hasn't moved, since $v_y = 0$, so $\Delta y = 0$ still. Since $v_y \propto \Delta y$, the velocity remains zero. The stone hangs in the air. If $v_y \propto t$, the stone's velocity is again zero at $t = 0$, but when we go one small moment forward, t increases. That means the velocity increases, and the stone begins moving.

2.1:



2.2:



2.3:



3.1: 5 m. This is the third period of 0.45 s, so the object should fall 5 times as far as it did in the first period.

3.2: 0.90 s. In the period from 0.45 s to 0.90 s the object falls 3 m, so it's fallen $1\text{ m} + 3\text{ m} = 4\text{ m}$ total.

3.3: 0.64 s, as we'll see soon. The object falls 1 m in 0.45 s and 4 m in 0.90 s, so your estimate should be between those values.

4.1: $20\text{ m} : 2.01\text{ s} \quad 80\text{ m} : 4.02\text{ s} \quad 320\text{ m} : 8.05\text{ s} \quad 1280\text{ m} : 16.1\text{ s}$

4.2: When we multiply the distance fallen by 4, we multiply the time it takes by 2. This is because $\Delta y \propto t^2$ and $4 = 2^2$.

4.3: $(n+1)^2 - n^2 = n^2 + 2n + 1 - n^2 = 2n + 1$. If we are at row n in the table in the solution to problem 4.1, in the next row, we will add $(2n+1)d$ as the next incremental distance. Since we are adding this to n^2d , our algebra shows the result will be $(n+1)^2d$, so this is an alternative way to show that the pattern in the table continues to hold for as many rows as we like.

4.4: By a factor of 4. (Play the clip back at $1/4$ speed.) That way the clip will take 4 times as long to play. When we multiply time by 4, the rule $\Delta y \propto t^2$ says we multiply free fall distance by 16, so the clip will look realistic.

4.5: Let's first set the period between distance measurements to be 1 s. In 1 s, an object with constant acceleration travels a distance d_1 . Then it should travel $2d_1$ between 1 s and 2 s, and $3d_1$ and $4d_1$ in the next two seconds, according to da Vinci. The total distance fallen in 4 s is $d_1 + 2d_1 + 3d_1 + 4d_1 = 10d_1$. Now let's repeat the analysis, but use a period of 2 s between distance measurements. Let the distance traveled in this much time be d_2 . Then in the next two seconds, it should travel $2d_2$. At this point, 4 s have passed and the object has traveled a total distance $d_2 + 2d_2 = 3d_2$. Finally, we compare these two predictions at 2 s and 4 s. In the first analysis, the object traveled $3d_1$ at 2 s and $10d_1$ at 4 s, so at 4 s it had traveled $\frac{10}{3}$ times as far as after 2 s. In the second analysis, it traveled 3 times as far after 4 s as after 2 s. We are analyzing the same motion each time, just with different time periods between distance measurements. We get two different values for the ratio of distances traveled between 2 s and 4 s, but at most one of these can be correct. da Vinci's hypothesis makes two *different* predictions for the same observation, so it is not only wrong, but internally inconsistent.